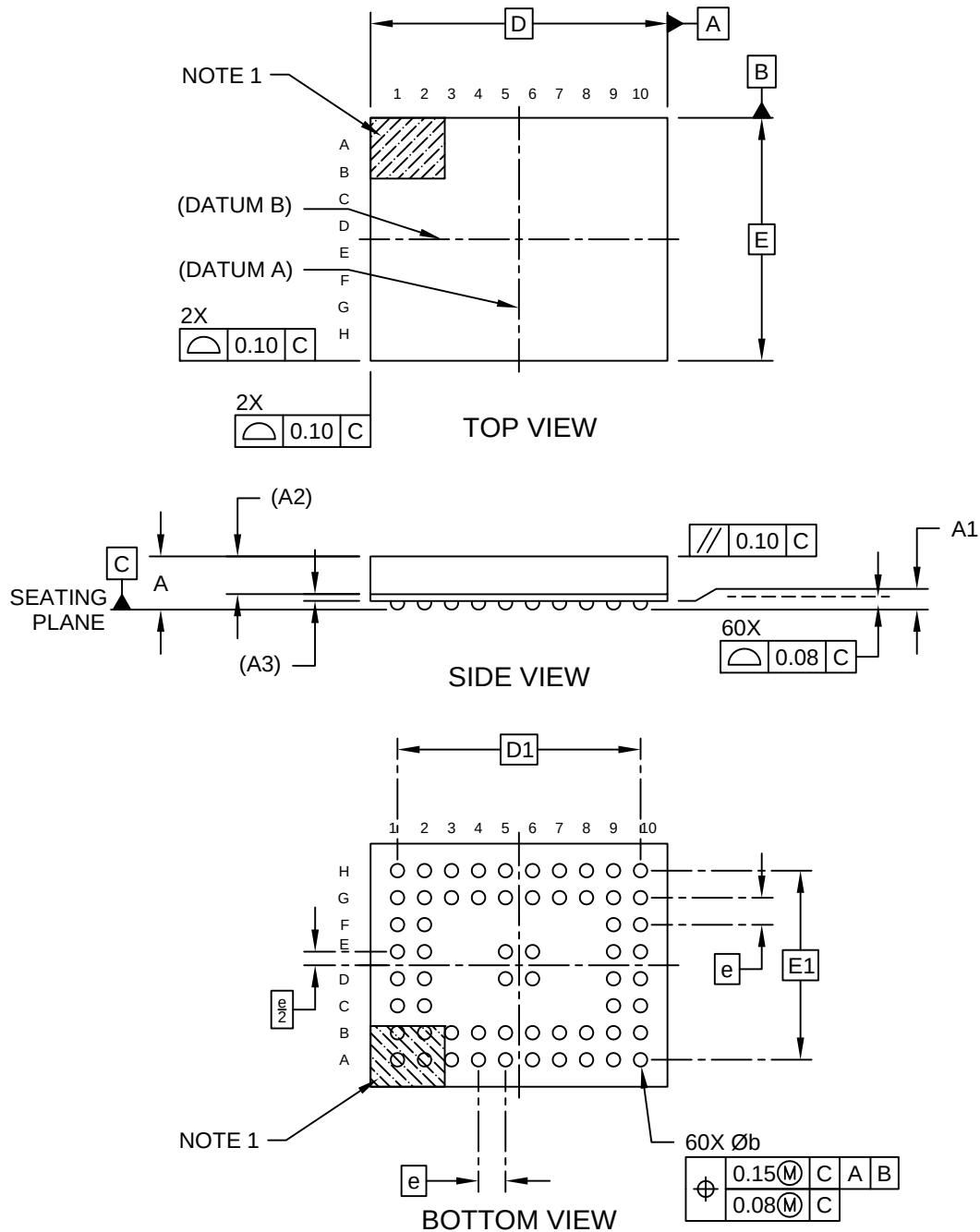


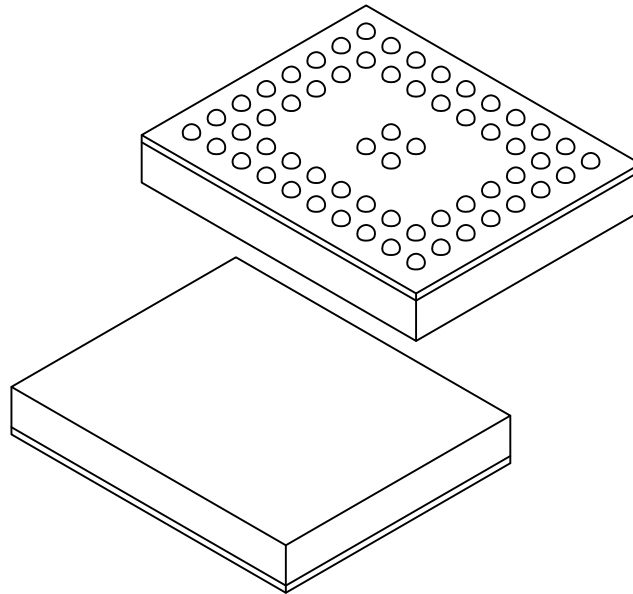
60-Ball Thin Fine Pitch Ball Grid Array (KMX) - 5.5x4.5 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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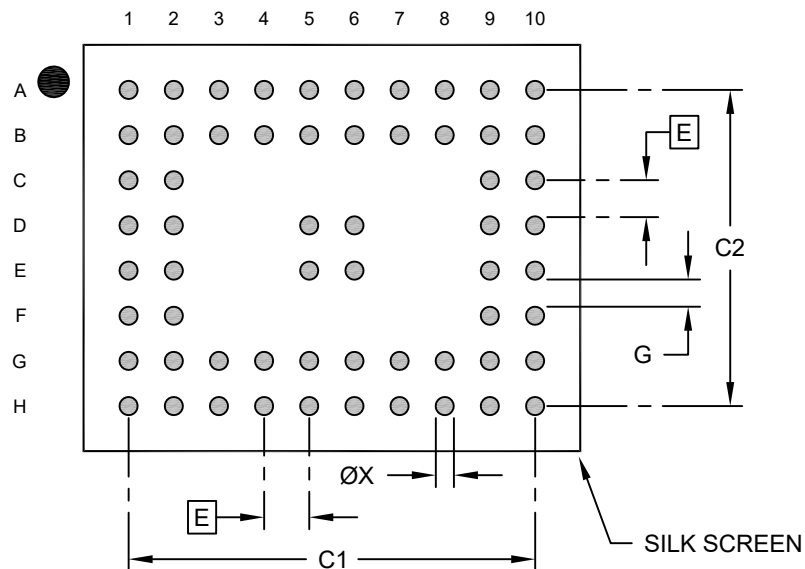
		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		60		
Pitch	e		0.50 BSC		
Overall Height	A		-	-	1.10
Standoff	A1		0.11	-	0.21
Mold Thickness	A2		0.70 REF		
Substrate Thickness	A3		0.125 REF		
Overall Length	D		5.50 BSC		
Overall Terminal Spacing	D1		4.50 BSC		
Overall Width	E		4.50 BSC		
Overall Terminal Spacing	E1		3.50 BSC		
Terminal Diameter	b		0.20	0.25	0.30

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

60-Ball Thin Fine Pitch Ball Grid Array (KMX) - 5.5x4.5 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Overall Contact Pad Spacing	C1		4.50	
Overall Contact Pad Spacing	C2		3.50	
Contact Pad Width (X60)	X			0.20
Contact Pad to Contact Pad	G	0.20		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2464 Rev. A